Game Engine PAL: Workplan, Weeks 1-4

Team of 3 · Plan start 10/14/2026 · Project deadline 3/10/2027 · Source: Game Engine PAL capstone proposal

This plan covers weeks 1-4 of the proposal only and sets up the work in weeks 5-8. Dates assume week 1 starts 10/14, so the Week 4 mentor review lands around 11/10.

Timeline check

The proposal is a 32-week plan, but 10/14/2026 to 3/10/2027 is about 21 weeks. Weeks 1-4 fit as written, and every later phase will have to be compressed by roughly a third. The proposal's upstream dates (Ada PAL release in Week 10, ECS PAL in Week 12) also need real calendar dates from the mentor.

Roles (even split, one deliverable per person per week)

- **A: Engine engineer.** Loop, subsystem boundaries, asset pipeline.
- **B: Game logic engineer.** Game design, entities, state machine.
- **C: Platform engineer.** Alire and CI, stub interfaces, embedded.

The proposal says the engine engineer and the game logic engineer must not be the same person.

Week 1 (10/14 - 10/20): setup and boundaries

Who	Task	Done when
A	Subsystem boundary map v0: what the engine provides versus Ada PAL versus ECS PAL	One-page diagram plus a package list
B	Game concept one-pager: genre, entity types, win and lose conditions	Written and agreed by the team
C	Alire workspace: `alire.toml` declaring Ada PAL and ECS PAL stubs, and CI running `alr build` on every commit	CI is green on `main`
All	Read the Design Rules and agree on conventions. Enforce LF line endings in `.gitattributes`, since `-gnatyd` rejects CRLF	Conventions in the repo

Week 2 (10/21 - 10/27): loop skeleton and stubs

Who	Task	Done when
A	`Frame_Duration` fixed-point type, `Loop_State` with the accumulator invariant, and a `Tick` skeleton using `Ada.Real_Time`, with no float time	Compiles on desktop
B	Entity and component catalogue and the state machine (menu, gameplay, game over)	Diagram plus table
C	Stub interfaces at the boundary for Ada PAL (present, input) and ECS PAL (entity and component ops). Pick the embedded board and confirm a ZFP runtime exists for it	Stubs build. Board chosen

Week 3 (10/28 - 11/3): tests, requests and sizing

Who	Task	Done when
A	Unit tests for the scheduler: clamp, bounded accumulator, interpolation factor between 0 and 1. First GNATprove run on `Game.Loop`, to see what already proves	Tests pass. Proof gaps listed
B	List of what game logic needs from the engine (spawn, input query, draw request), written as requests to A. No PAL imports	Interface request list
C	Asset pipeline check: the `qoi` crate builds on desktop, with embedded feasibility notes. First ECS storage sizing from B's catalogue. Contact the Ada PAL and ECS PAL teams on release dates and gaps	Sizing table. Release dates written down

Week 4 (11/4 - 11/10): architecture review

Who	Task	Done when
A	Architecture document draft (loop, boundaries, asset pipeline) and lead the review	Draft circulated 2 days ahead
B	Scope for weeks 9-12 (first playable slice) and a boundary-rule self-check of the game-side code	Scope and risk list
C	Evidence pack: green `alr build`, stub and dependency status, embedded risks	One-page status report
All	**Mentor architecture review.** Get `Frame_Duration` approved, because the proposal treats any later float time as structural	Action items recorded

Exit criteria for week 4

- `alr build` is green on every commit.
- The loop skeleton runs with fixed-point time and tests.
- The boundary map is approved by the mentor, with `Frame_Duration` signed off.

- The stub interfaces are in place for weeks 5-8: loop ticking against present, ECS wired, QOI loading, and input routed.
- The Ada PAL and ECS PAL release dates are written down.

Risks

- The repo already has seed code (`yarlib`, `escape`). Confirm with the mentor whether these are the "stub versions" the proposal means, or whether you should depend on the real Ada PAL and ECS PAL crates.
- `escape` has milestones that aren't finished yet, so ECS operations can't be fully used. That's why the plan uses stubs at the boundary.
- If the Week 4 review slips, the weeks 5-8 work (loop against present, ECS wiring) starts without an approved type.